House Price Prediction Using Machine Learning

Project Overview

The objective of this project was to develop a machine learning model capable of predicting house prices based on various property characteristics. A real-world housing dataset was used, containing features such as area, number of bedrooms, bathrooms, stories, parking spaces, furnishing status, and other amenities. The project involved data preprocessing, exploratory analysis, model development, performance evaluation, and result visualization.

Data Preprocessing

The dataset was loaded using Pandas and examined for missing values and duplicate records. Duplicate entries were removed, and categorical variables were converted into numerical format using one-hot encoding. This preprocessing step ensured that the data was suitable for machine learning algorithms.

Model Development and Evaluation

Two regression models were developed and compared:

1. Linear Regression
2. Random Forest Regressor

The dataset was divided into training and testing sets using an 80:20 ratio. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Results

Model	**MAE**	**RMSE**	**R^2 Score**
Linear Regression	970,043	1,324,507	0.653
Random Forest Regressor	1,021,546	1,400,566	0.612

The results indicate that Linear Regression performed better than Random Forest for this dataset, achieving the lowest prediction error and the highest R^2 score.

Key Findings

Feature importance analysis revealed that the area of the house was the most influential factor affecting price, followed by bathrooms, air conditioning, parking availability, and number of stories. Larger houses with better amenities generally had higher market values.

Conclusion

This project successfully demonstrated the use of machine learning techniques for house price prediction. The Linear Regression model provided the best overall performance, explaining approximately 65.3% of the variation in house prices. The findings highlight the importance of property size and essential amenities in determining house values and can support data-driven decision-making in the real estate industry.